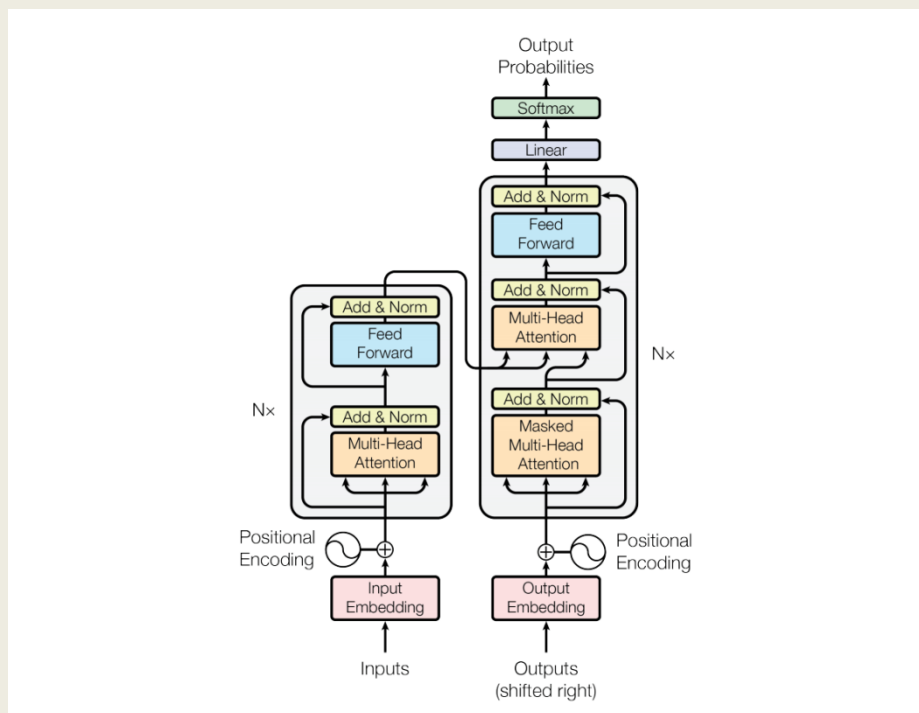




NLP Transformer-based Models used for Sentiment Analysis



BERT (Bidirectional Encoder Representations from Transformers): A powerful language model that learns deep bidirectional representations by jointly conditioning on both left and right context in all layers. It revolutionized NLP tasks like question answering and text classification.

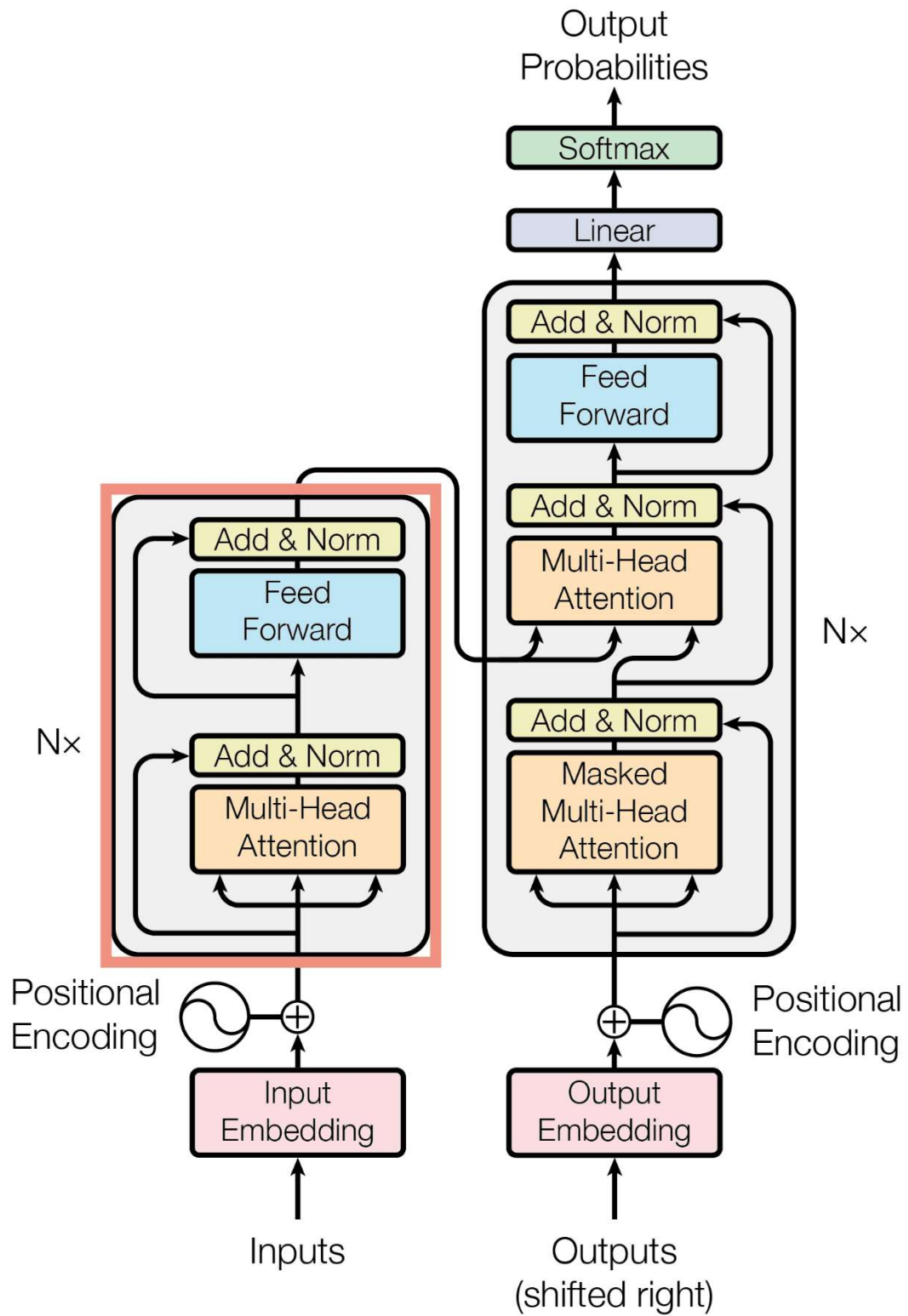
RoBERTa (Robustly Optimized BERT Approach): An improved version of BERT that focuses on optimizing the training process. It uses larger datasets, longer training times, and different training techniques to achieve better performance than the original BERT.

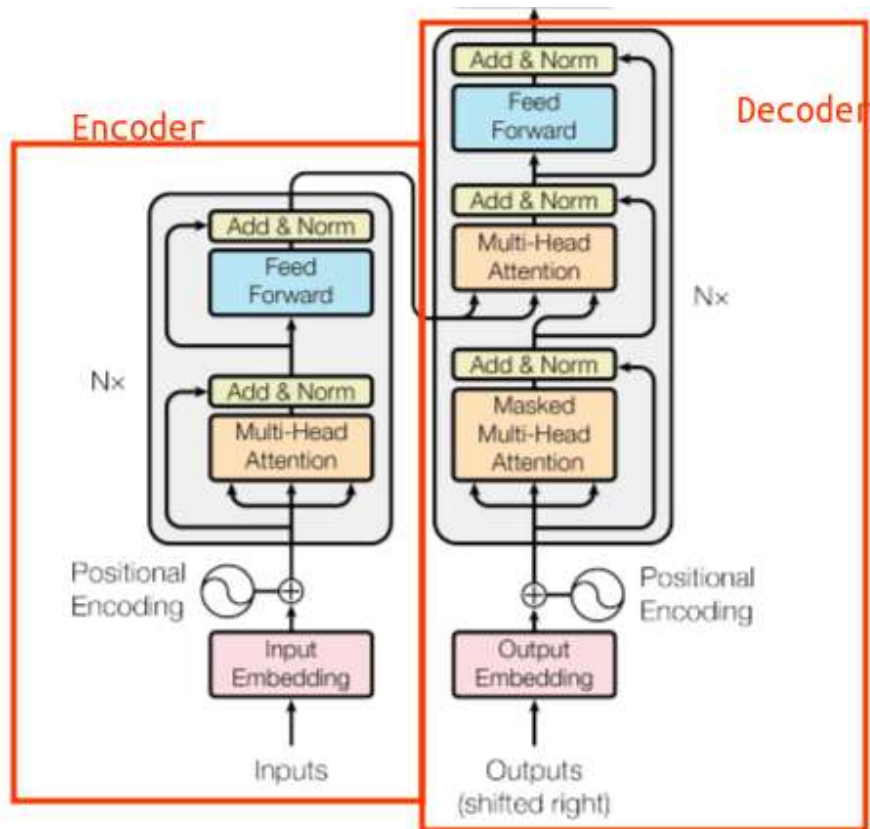
DistilBERT: A smaller, faster, and cheaper version of BERT that is distilled from a larger BERT model. It retains most of the performance of the original BERT while being significantly more efficient.

ALBERT (A Lite BERT for Self-supervised Learning of Language Representations): A BERT variant that reduces the model size and parameter count by factorizing the parameter matrix and sharing parameters across layers. This makes it more efficient and faster to train.

XLNet: A generalized autoregressive pretraining method that learns bidirectional contexts by maximizing the likelihood of a sequence of tokens in a non-autoregressive manner. It outperforms BERT on many NLP tasks, especially those requiring understanding long-range dependencies.

Transformer Architecture





The transformer architecture is designed to process sequential data, such as natural language, in a highly efficient and effective manner. Unlike traditional models that rely on sequential processing (like RNNs), transformers utilize a mechanism called self-attention, allowing them to analyze the entire input sequence at once. This capability enables them to capture complex relationships and dependencies between tokens in the sequence.

The transformer model consists of two main parts:

- 1. Encoder:** The encoder processes the input sequence and generates a continuous representation of it. This representation captures the contextual information of the input tokens.
- 2. Decoder:** The decoder takes the encoder's output and generates the final output sequence. It does this by predicting one token at a time, using the encoded representations and previously generated tokens.

Both the encoder and decoder are composed of multiple identical layers—typically six layers in the original transformer architecture—allowing for deep learning and complex feature extraction.

Key Components of Transformers

1. Multi-Head Attention: